

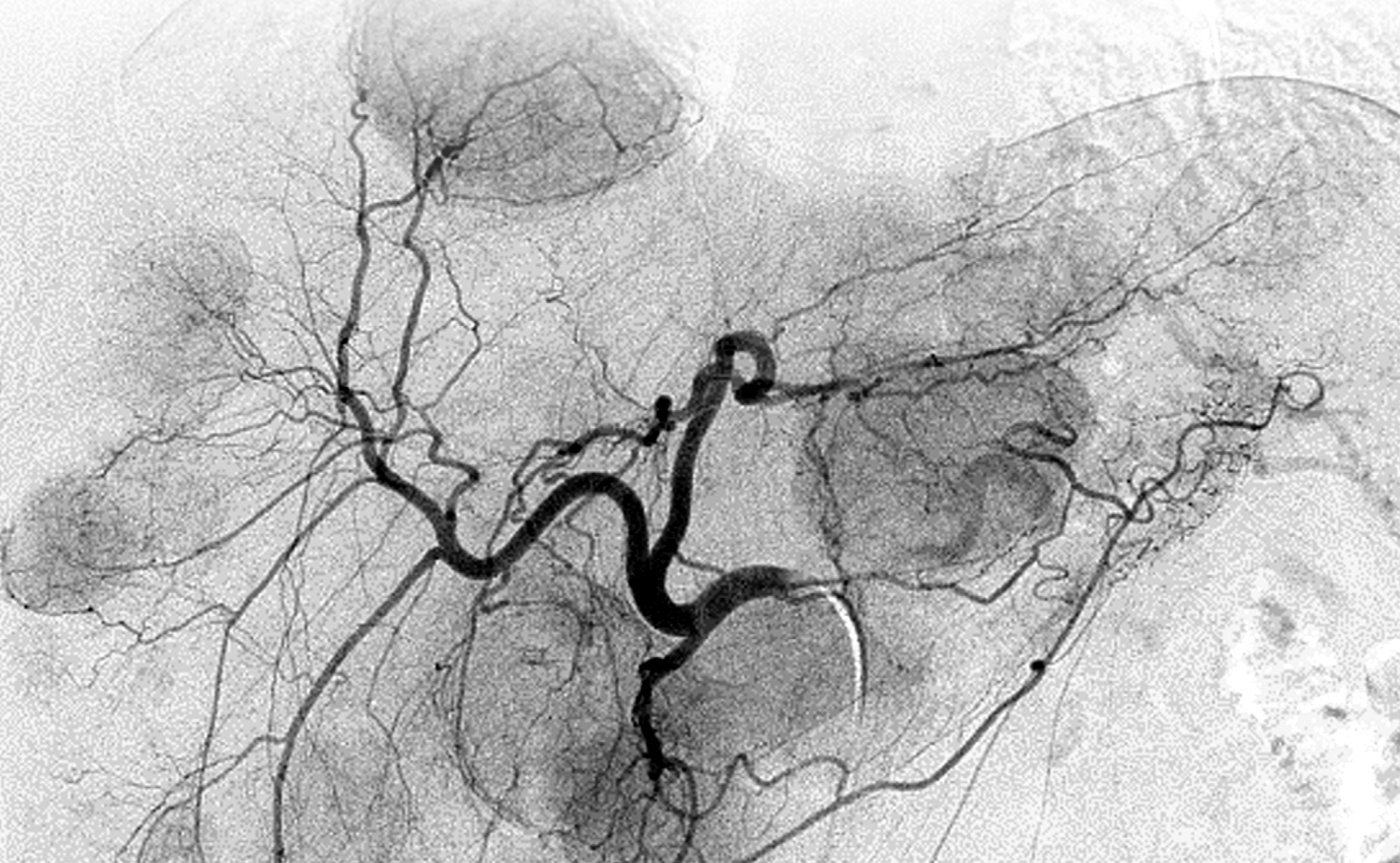
Hepatocellular Adenomas: Molecular Basis and Multimodality Imaging Update

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Hepatocellular adenomas (HCAs) are a family of liver tumors that are associated with variable prognoses. Since the initial description of these tumors, the classification of HCAs has expanded and now includes eight distinct genotypic subtypes based on molecular analysis findings. These genotypic subtypes have unique derangements in their cellular biologic makeup that determine their clinical course and may allow noninvasive identification of certain subtypes. Multiphasic MRI performed with hepatobiliary contrast agents remains the best method to noninvasively detect, characterize, and monitor HCAs. HCAs are generally hypointense during the hepatobiliary phase; the β -catenin–mutated exon 3 subtype and up to a third of inflammatory HCAs are the exception to this characterization. It is important to understand the appearances of HCAs beyond their depictions at MRI, as these tumors are typically identified with other imaging modalities first. The two most feared related complications are bleeding and malignant transformation to hepatocellular carcinoma, although the risk of these complications depends on tumor size, subtype, and clinical factors. Elective surgical resection is recommended for HCAs that are persistently larger than 5 cm, adenomas of any size in men, and all β -catenin–mutated exon 3 HCAs. Thermal ablation and transarterial embolization are potential alternatives to surgical resection. In the acute setting of a ruptured HCA, patients typically undergo transarterial embolization with or without delayed surgical resection. This update on HCAs includes a review of radiologic–pathologic correlations by subtype and imaging modality, related complications, and management recommendations.

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Supplemental Material



Quiz questions for this article are available through the Online Learning Center.

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Abbreviations: FDG = fluorodeoxyglucose, FNH = focal nodular hyperplasia, HCA = hepatocellular adenoma, HCC = hepatocellular carcinoma, HNF-1 α = hepatocyte nuclear factor 1 α inactivated, OATP = organic anion transporting polypeptide, OC = oral contraceptive

TEACHING POINTS

- At MRI, inflammatory HCAs show moderate T2 hyperintensity corresponding to dilated sinusoids. In addition, between 24% and 54% of these tumors exhibit the atoll sign, with which there is a peripheral rim of higher T2 signal intensity that may have delayed enhancement.
- Decreased L-FABP (leading to intracellular fat) and decreased OATP1B1/3 expression in HNF-1 α HCAs cause decreased signal intensity at T1-weighted opposed-phase and hepatobiliary phase MRI, respectively.
- An important distinguishing feature of β -catenin–mutated exon 3 HCAs is that most (>80%) are iso- or hyperintense during the hepatobiliary phase, corresponding to increased OATP1B1/3 expression.
- Independent risk factors for symptomatic bleeding include HCA subtype (β -catenin–mutated exon 7/8 and sonic hedgehog), chronic alcohol consumption, and larger tumor size.
- β -catenin–mutated exon 3 HCA subtype, tumor size larger than 5 cm, male sex, and chronic liver disease (specifically, glycogen storage disorders and vascular liver disease) are established risk factors for HCC transformation.

Introduction

Hepatocellular adenomas (HCAs) are a heterogeneous group of liver tumors associated with variable prognoses. When first discovered in the 1970s, HCAs were believed to reflect a single entity (1). Over time, these tumors have been stratified into an increasing number of unique genotypic subtypes based on molecular analysis findings. The first major update was in 2006, when the Bourdeaux group described four HCA subtypes (2,3): inflammatory, hepatocyte nuclear factor 1 α inactivated (HNF-1 α), β -catenin–mutated exon 3, and unclassified. The second major update occurred in 2017, when four additional groups were added: sonic hedgehog, β -catenin–mutated exon 7/8, mixed inflammatory and β -catenin–mutated exon 3, and mixed inflammatory and β -catenin–mutated exon 7/8 (4). With the exception of mixed inflammatory and β -catenin–mutated subtypes, HCA subtypes develop from mutually exclusive molecular pathways, with distinct risk factors and prognoses (5).

HCAs classically occur in women of childbearing age who have engaged in long-term use of oral contraceptives (OCs), the latter of which increases the incidence of these tumors 30- to 40-fold (6). All HCA subtypes have a female predilection (60%–100%), and median OC use ranges from 8 to 22 years (4). Other risk factors include anabolic androgenic steroid use, obesity, alcohol consumption, and certain genetic syndromes. Although HCAs were initially believed to be rare, the incidence of these tumors has increased over time. This may be due to a combination of increased use of cross-sectional imaging, increasing rates of obesity, and increased anabolic substance use (6,7). HCAs can be solitary or multiple, and adenomatosis (more than 10 adenomas) has been described with all subtypes (4,8). In patients with multiple HCAs, the

molecular subtypes tend to be the same; however, the mutations at the codon level are not (4).

Rates of complications, which include bleeding and malignant transformation, depend on the HCA subtype (4,5). Common HCA subtypes have variable imaging appearances and may mimic other liver neoplasms (6,9,10). Thus, it is important to accurately identify, characterize, and distinguish HCAs when they are encountered in clinical practice. The reference standard for determining the HCA subtype is either immunohistochemical stain analysis or rarely, genomic sequencing with use of a surgical pathologic specimen (4,5). However, such tests may be unavailable outside of a tertiary liver center or unfamiliar to the pathologist. Also, it is impractical to resect or biopsy every suspected HCA lesion, particularly if the patient has adenomatosis. The HCA subtype can often be suggested by the imaging findings, obviating the need for an invasive procedure.

In this article, we review the clinical-epidemiologic, imaging, and histopathologic features; complications; and management of HCA subtypes on the basis of the most recent molecular classification systems.

Radiologic-Pathologic Correlation by Molecular Subtype

Inflammatory HCA

Inflammatory HCA is tied with HNF-1 α HCA as the most common subtype (30%–35% of HCAs) and is strongly associated with higher body mass index and OC use. In patients with inflammatory HCA, the median body mass index is 25 kg/m² and the median length of OC use is 20 years (4). Other risk factors include alcohol use, a fatty liver, and glycogen storage disorders. Patients may have inflammatory syndrome (fever, elevated white blood cell count, elevated fibrinogen level, and elevated C-reactive protein level) or abnormal liver test results (commonly elevated alkaline phosphatase and γ -glutamyl transferase concentrations) (2,4–6,11).

At the molecular level, inflammatory HCAs have sustained activation of the interleukin 6/Janus kinase/signal transducer and activator of transcription (STAT) pathway through somatic gain-of-function mutations (11). At the histologic level, these HCAs can exhibit patchy inflammatory infiltration, ductular reaction, sinusoidal dilatation, and dystrophic blood vessels (pseudoportal tracks with thick-walled arteries and lack of veins or bile ducts). Inflammatory infiltration is responsible for diffuse strong serum amyloid A and C-reactive protein immunohistochemical staining, a hallmark of inflammatory HCAs at immunohistochemical analysis (5,11–13). Sinusoidal dilatation and dystrophic blood vessels are responsible for persistent enhancement during the portal venous phase of iodinated or gadolinium-based contrast agent-enhanced CT or MRI, respectively. Dilated sinusoids are also believed to be responsible for the “atoll” sign and T2 hyperintensity (14–16); thus, these HCAs were once termed *telangiectatic focal nodular hyperplasia* (FNH) until they were reclassified as inflammatory HCA by the World Health Organization (5,16). At histopathologic analysis, the surrounding liver can also exhibit sinusoidal dilatation and steatosis, which are present in up to 40% of patients (4,5,13,16,17).

At MRI, inflammatory HCAs show moderate T2 hyperintensity corresponding to dilated sinusoids. In addition, between 24% and 54% of these tumors exhibit the atoll sign, with which there is a peripheral rim of higher T2 signal intensity that may have delayed enhancement (15,18,19). Analogous to an atoll, the peripheral rim represents the coral, the HCA represents the lagoon, and the background liver represents the open ocean. Although it has variable sensitivity, the atoll sign is highly specific (97%) for the inflammatory subtype (18). The atoll sign can also be seen on gray-scale US images and depicted as a rim of delayed enhancement on contrast-enhanced US images (19). Rarely, inflammatory HCAs have microscopic fat on T1-weighted chemical shift MR images. The background liver may show steatosis (14,20,21).

Inflammatory HCAs have variable T1 signal intensity relative to the background liver, which may depend on background hepatic steatosis. During dynamic postcontrast imaging, these tumors have avid and peak enhancement during the arterial phase that persists above the background liver during the portal venous phase. In the hepatobiliary phase, most inflammatory HCAs are hypointense to the background liver (18,20,21). Figure 1 shows a large (>5 cm) typical inflammatory HCA with an atoll sign and subsequent rupture. However, it is recognized that a small proportion of the tumor may be iso- or hyperintense relative to the liver during the hepatobiliary phase (10). In some cases, the increased HCA signal intensity in the hepatobiliary phase can be attributed to preserved or increased organic anion transporting polypeptide (OATP) expression (20,22), while in other cases, it may be due to background hepatic steatosis causing relative HCA hyperintensity after fat suppression (23). Inflammatory HCAs with true hepatobiliary phase uptake (ie, tumor isointensity to hyperintensity [relative to the liver] that is not attributed to background steatosis) are a potential pitfall in distinguishing this HCA subtype from other liver tumors (9,10). Figure 2 shows an inflammatory HCA with hepatobiliary phase isointensity.

At US, the echogenicity of inflammatory HCAs relative to the liver varies and may depend on the degree of hepatic steatosis (19,24–26). Figure 3 shows how an HCA may appear hypoechoic when the liver has background steatosis. With contrast-enhanced imaging, venous washout is more commonly observed at contrast-enhanced US than at contrast-enhanced MRI or CT (19,26,27). This discrepancy can be attributed to the diffusion of gadolinium- or iodine-based contrast agents into the disordered interstitial spaces, whereas contrast-enhanced US involves the use of purely intravascular contrast agents.

HNF-1 α HCA

HNF1A is a tumor suppressor gene that also acts as a major hepatocyte transcription factor. It is responsible for the expression of liver fatty acid-binding protein (L-FABP) and organic anionic transporter proteins (5,27,28). L-FABP is responsible for fatty acid trafficking, and the downregulation of this protein leads to accumulation of intracellular fat. At histopathologic analysis, macrovesicular fat and lack of L-FABP staining at immunohistochemical analysis are hallmarks of HNF-1 α HCA (5,12).

Decreased L-FABP (leading to intracellular fat) and decreased OATP1B1/3 expression in HNF-1 α HCAs cause decreased signal intensity at T1-weighted opposed-phase and hepatobiliary phase MRI, respectively. Most (90%) HNF-1 α HCAs have acquired somatic mutations, while the remaining 10% of these tumors have germline and somatic mutations. For example, patients with maturity-onset diabetes of the young–3 (MODY 3) already have at least one congenital preexisting mutation and thus are at a higher risk for HNF-1 α HCA (5). A patient may have hepatocellular adenomatosis, and the HCAs within the same patient can have either shared or separate independent mutations (4). The association between HNF-1 α HCA and either OC use or obesity is not as strong as that between inflammatory HCA and these factors. In patients with HNF-1 α HCA, the median length of OC use is 12 years and the median body mass index is 21 kg/m² (4). HNF-1 α HCAs have not been reported among patients with glycogen storage diseases (29,30). Background fatty liver is less common with this HCA subtype (3,4,14,20).

At MRI, HNF-1 α HCAs are readily identified by a loss of signal at T1-weighted opposed-phase imaging, which is due to intracellular microscopic fat (14,16,18,20,21). The presence of microscopic fat is highly sensitive and specific for HNF-1 α HCA, with a contemporary series (18) reporting a sensitivity of 95%, specificity of 85%, and area under the receiver operating characteristic curve of 90%. Intratumoral fat can be quantified, and HNF-1 α HCAs have an average fat fraction of 24% (21). HNF-1 α HCAs have variable signal intensity at unenhanced T1-weighted fat-suppressed MRI (14,16) but may appear hypointense relative to the liver owing to the presence of fat (3,21). Like other HCA subtypes, HNF-1 α HCAs show peak enhancement during the arterial phase (14,16,18–22). However, unlike other HCA subtypes, they do not show hyperenhancement during the venous phase (16) and may be hypoenhancing, giving the appearance of washout (20,21). In the hepatobiliary phase, HNF-1 α HCAs are markedly hypointense and, compared with other subtypes, have the lowest signal intensity relative to the background liver owing to reduced OATP1B1/3 expression (18,20–22). Figure 4 shows typical MRI and US images depicting HNF-1 α HCA.

At CT, HNF-1 α HCAs can have appearances similar to their appearances on dynamic MR images, including hypoattenuation in the unenhanced phase and peak enhancement in the arterial phase. While MRI is much more sensitive for the presence of intratumoral fat, unenhanced CT may occasionally show areas of low attenuation corresponding to fat (Fig 5). Investigators in one study of unsubtyped HCAs (31) identified fat in 7% of cases, while the prevalence of HNF-1 α HCAs among the recent above-mentioned research cohorts is closer to 30%, suggesting that CT would underestimate HNF-1 α HCAs.

At fluorodeoxyglucose (FDG) PET/CT, HNF-1 α HCAs can be intensely FDG avid and mimic malignancy, with the maximal standardized uptake value (SUV_{max}) reported to be as high as 12.5 (Fig 6) (32). This may be due to overexpression of glucose transporter type 1 in an HNF-1 α HCA rather than true hypermetabolic activity (33,34). Although fatty changes can induce an inflammatory response and cause hypermetabolic activity, in a case series, Lee et al (32) did not identify inflammatory changes in HNF-1 α HCAs at histopathologic

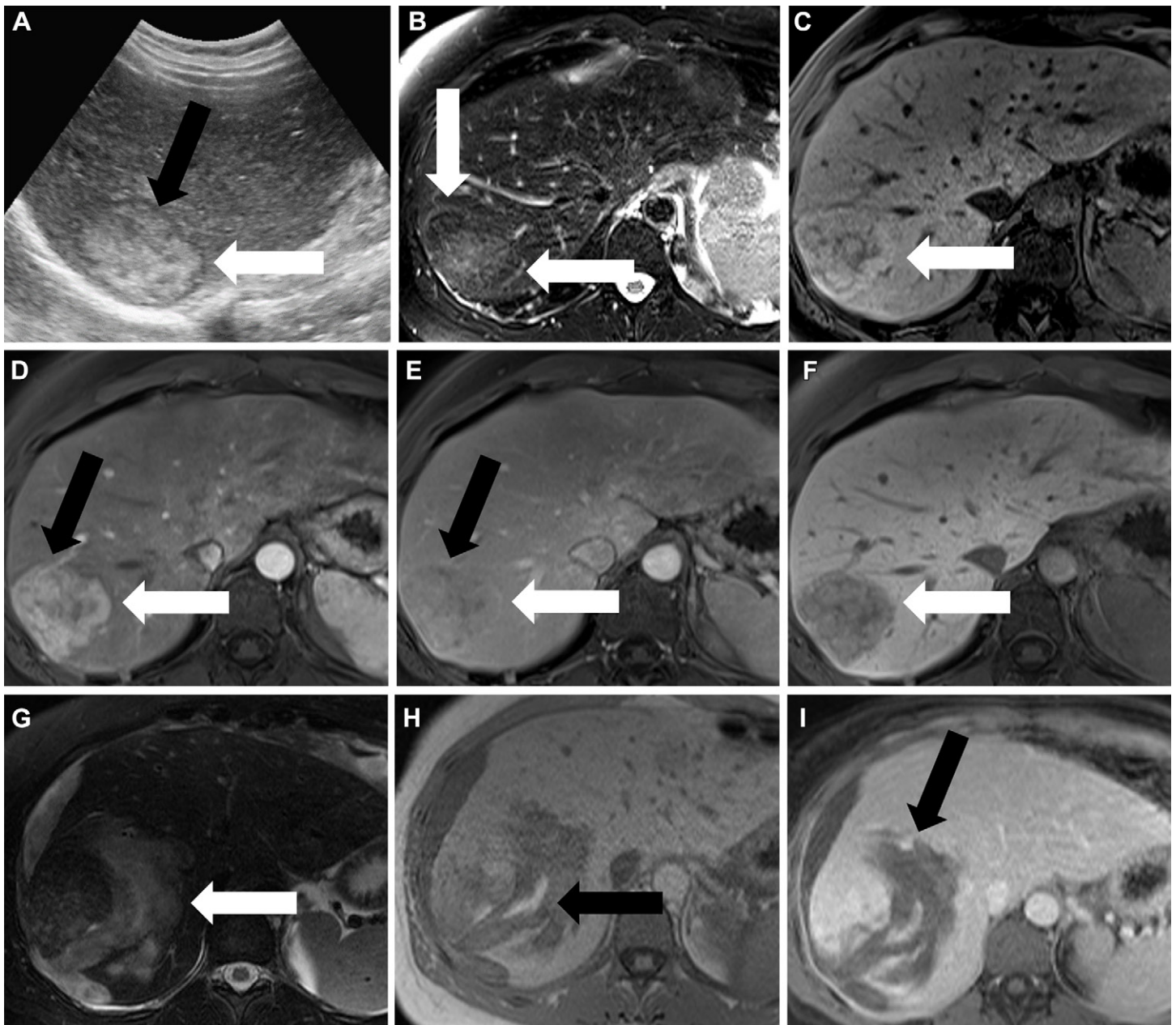


Figure 1. Large (>5 cm) surgically proven inflammatory HCA in the right hepatic lobe in a 36-year-old woman. (A) Transabdominal gray-scale US image shows a hyperechoic mass (white arrow) in the posterior right hepatic lobe, with a hypoechoic rim (black arrow) corresponding to the atoll sign. (B) Axial T2-weighted fat-suppressed MR image shows a hyperintense mass (arrows) that has a rim of higher signal intensity corresponding to the atoll sign. (C–F) Axial dynamic gadoxetic acid-enhanced T1-weighted gradient-recalled-echo fat-suppressed MR images show the mass (white arrow) isointense to slightly hyperintense during the unenhanced phase (C); hyperenhancing during the hepatobiliary phase, with a rim of hypoenhancement (black arrow) corresponding to the atoll sign (D); isoenhancing during the portal venous phase, with a hyperenhancing rim (ie, atoll sign) (black arrow) (E); and hypointense compared with the liver during the hepatobiliary phase (F). (G) One year later, the patient presented with right upper quadrant pain and symptoms of hypotension and underwent emergent MRI. Axial T2-weighted fat-suppressed MR image shows heterogeneous fluid signal intensity (arrow) within the HCA and throughout the peritoneum. (H) Axial extracellular gadolinium-enhanced T1-weighted fat-suppressed MR image during the unenhanced phase shows heterogeneously high intrinsic signal intensity (arrow) corresponding to hemorrhage. (I) Axial portal venous phase T1-weighted MR image shows active bleeding (arrow). The patient underwent emergent right hepatectomy.

analysis. Furthermore, HNF-1 α HCAs are reported to have a higher SUV_{max} than inflammatory HCAs (27).

At US, most HNF-1 α HCAs are hypervascular and homogeneously hyperechoic owing to the presence of intracellular fat (Fig 4) (19,35). Although the relative echogenicity can be affected by background or “nontumoral” hepatic steatosis, in two US series (19,35), hepatic steatosis was rarely identified

among HNF-1 α HCAs but commonly identified among inflammatory HCAs. Clinically, HNF-1 α HCAs, unlike inflammatory HCAs, do not have a strong association with obesity (4). At contrast-enhanced US, HNF-1 α HCAs have peak enhancement in the arterial phase and appear isoechoic to the liver during the portal venous and delayed phases (19). HNF-1 α HCAs may mimic hemangiomas at gray-scale US,

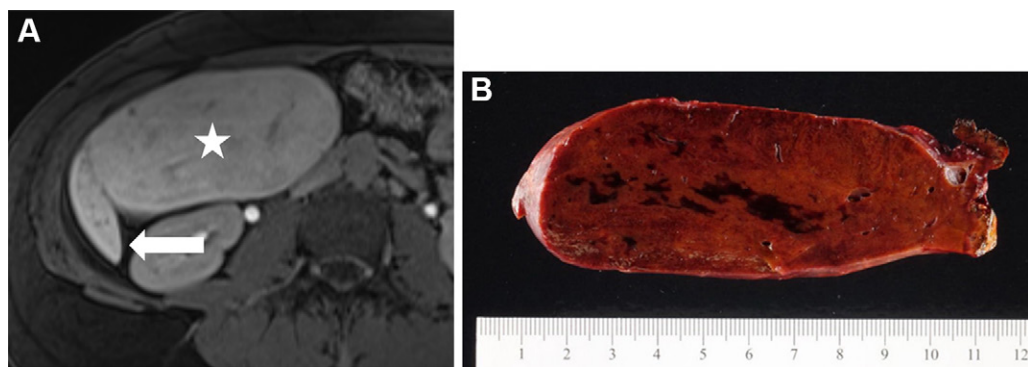


Figure 2. Large (>5 cm) surgically proven inflammatory HCA in the left hepatic lobe in a 35-year-old woman. (A) Axial gadoxetic acid-enhanced hepatobiliary phase T1-weighted fat-suppressed MR image shows an exophytic mass (☆) that is isointense to the background liver (arrow). (B) Gross specimen shows an encapsulated mass with central areas of hemorrhage.

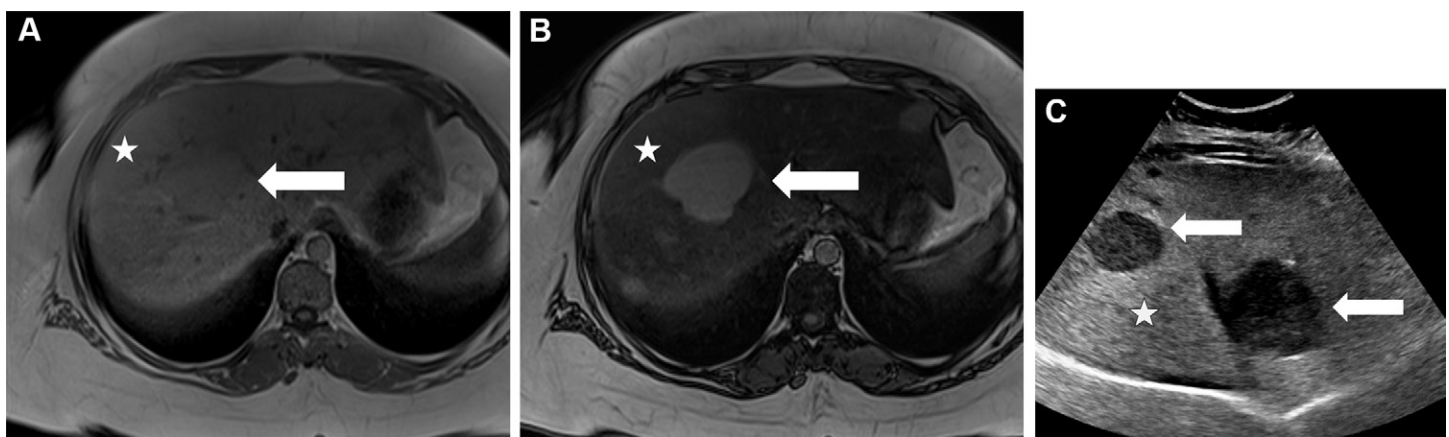


Figure 3. Metabolic syndrome and biopsy-proven inflammatory adenomatosis in a 22-year-old woman. (A, B) Axial in-phase (A) and opposed-phase (B) T1-weighted chemical shift MR images show diffuse signal loss in the liver parenchyma (☆), but not in the HCA (arrow), between the in phase and opposed phase. (C) On the transabdominal gray-scale US image, the HCAs (arrows) are hypoechoic compared with the background fatty liver (☆).

and multiphasic CT/MRI or contrast-enhanced US (36) may be required to distinguish between these two tumors (Fig S1).

Bleeding (Fig 5) and malignant transformation (Fig 6) are rare but well-recognized complications of HNF-1 α HCA, particularly when these tumors are large (>5 cm) (4). The molecular pathogenesis to hepatocellular carcinoma (HCC) is not known, but it is not related to the β -catenin pathway, which drives HCC transformation in β -catenin-mutated HCAs (5). Malignant transformation occurs in up to 4% of HNF-1 α HCAs (4). HNF-1 α HCA is the subtype most likely to progress in size or number despite the cessation of OC use and is over-represented in postmenopausal patients older than 60 years (Fig 6) (37,38). Some of these cases may reflect a newer “myxoid subtype,” a histologic descriptor based on extracellular myxoid material within the tumor. This myxoid subtype can be considered a subcategory within the HNF-1 α HCA subtype that is based on loss of L-FABP expression (12) and is associated with older age and malignant transformation (39,40). It is possible that some HNF-1 α HCAs are hormonally independent after development, and as a whole this subtype has the weakest association with OC use and obesity.

β -Catenin-mutated HCA

β -catenin-mutated HCAs are a family of HCAs that share a missense mutation in the cadherin-associated protein β 1 gene (*CTNNB1*) at exon 3 or exon 7/8 (4,5). This mutation leads to variably increased activation of the wntless-type integration site protein ligand/ β -catenin pathway, which regulates multiple cellular processes such as proliferation and differentiation (41). Aberrant activation of this pathway is shared among malignancies and is implicated in up to 35% of HCCs (42). *CTNNB1* mutations at exon 3 lead to strong pathway activation, and, thus, this β -catenin-mutated HCA subtype has the highest risk (as high as 40%) of HCC transformation (3,4). Exon 3 mutations are also associated with anabolic androgenic steroid use, vascular liver disease, and male sex, with 30%–40% of these mutations occurring in men (4).

On the other hand, exon 7/8 mutations involve a negligible (though nonzero) risk of HCC transformation and are highly associated, just after sonic hedgehog HCAs, with histologic and symptomatic bleeding (4,5,43). Exon 7/8 mutations were discovered relatively recently (4) and before the associated HCAs were included in the unclassified subtype (7,44). In one study (4), β -catenin-mutated exon 7/8 HCAs

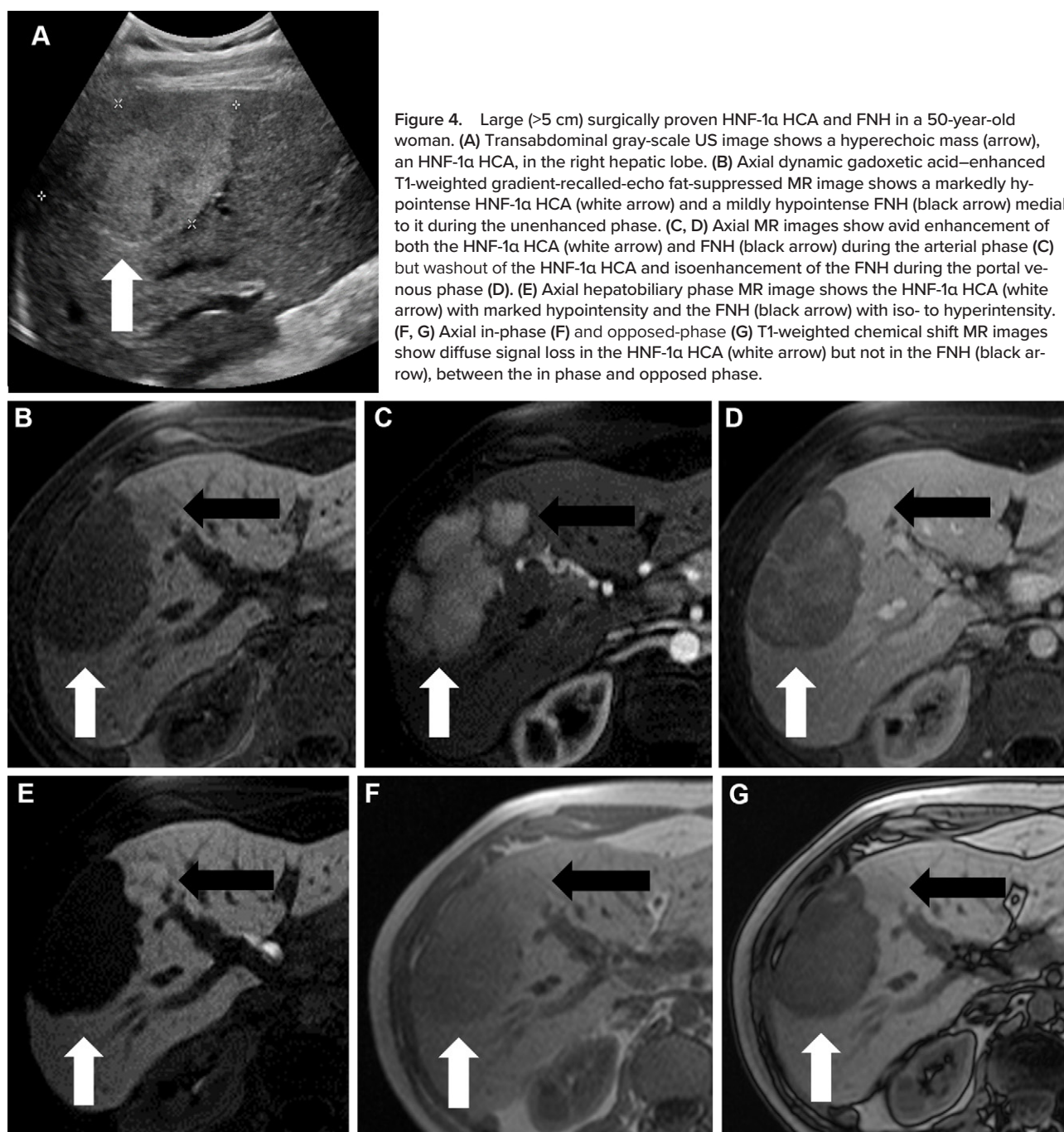


Figure 4. Large (>5 cm) surgically proven HNF-1 α HCA and FNH in a 50-year-old woman. (A) Transabdominal gray-scale US image shows a hyperechoic mass (arrow), an HNF-1 α HCA, in the right hepatic lobe. (B) Axial dynamic gadoxetic acid-enhanced T1-weighted gradient-recalled-echo fat-suppressed MR image shows a markedly hypointense HNF-1 α HCA (white arrow) and a mildly hypointense FNH (black arrow) medial to it during the unenhanced phase. (C, D) Axial MR images show avid enhancement of both the HNF-1 α HCA (white arrow) and FNH (black arrow) during the arterial phase (C) but washout of the HNF-1 α HCA and isoenhancement of the FNH during the portal venous phase (D). (E) Axial hepatobiliary phase MR image shows the HNF-1 α HCA (white arrow) with marked hypointensity and the FNH (black arrow) with iso- to hyperintensity. (F, G) Axial in-phase (F) and opposed-phase (G) T1-weighted chemical shift MR images show diffuse signal loss in the HNF-1 α HCA (white arrow) but not in the FNH (black arrow), between the in phase and opposed phase.

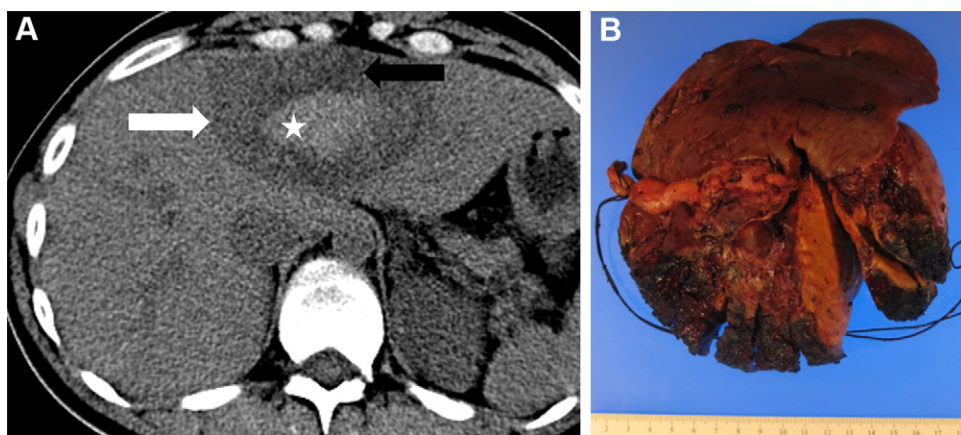


Figure 5. Large (>5 cm) surgically proven HNF1- α HCA in a 28-year-old woman. (A) Axial noncontrast CT image shows a mass (white arrow), an HNF1- α HCA, in the left hepatic lobe, with areas of low attenuation (black arrow) that correspond to microscopic fat, and a central area of high attenuation (☆) corresponding to histologic hemorrhage. (B) Gross pathologic specimen shows an exophytic mass with areas of hemorrhage.

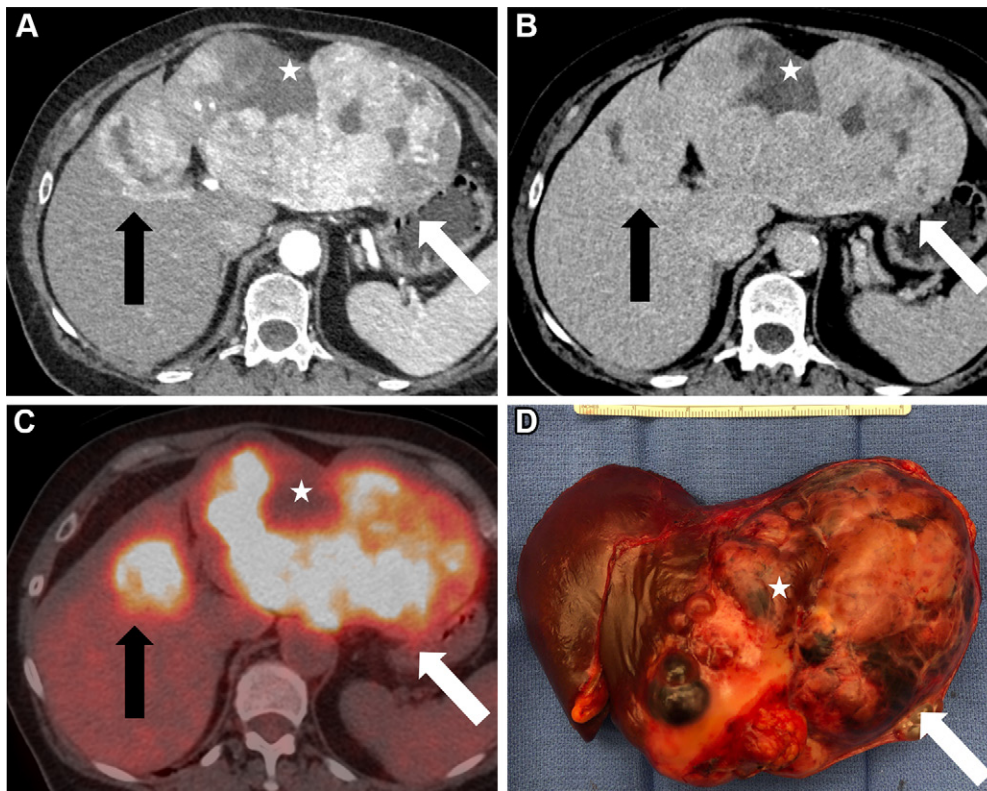


Figure 6. Two separate HNF-1 α HCAs, with hepatocellular carcinoma (HCC) arising from the larger HCA, in a 71-year-old postmenopausal woman. (A) Axial arterial phase CT image shows a mass (black arrow), an HCA, in the medial segment left hepatic lobe, and a larger more heterogeneous mass (white arrow), corresponding to HCC transformation (ie, HCA-HCC), replacing the lateral segment. (B) Axial delayed phase CT image shows subtle areas of washout in the HCA (black arrow) and HCA-HCC (white arrow). (C) On the axial FDG PET/CT image, both the HCA (black arrow) and the HCA-HCC (white arrow) are intensely hypermetabolic. The HCA-HCC has areas of necrosis (☆ in A–C) that correspond to the areas of nonenhancement at CT (A and B) and to photopenia at FDG PET/CT (C). (D) Gross pathologic specimen shows both masses (arrow) and the area of necrosis (☆).

had the highest proportion of all HCA subtypes associated with glycogen storage diseases (22%), while the newly defined unclassified group had none, suggesting redistribution with the new update (4). Figure S2 shows an example of a patient with glycogen storage disease and mixed inflammatory and β -catenin–mutated exon 7/8 adenomatosis.

A small proportion of inflammatory HCAs also harbor *CTNNB1* mutations at exon 3 or exon 7/8, resulting in four HCA subtypes that share this mutation: β -catenin–mutated exon 3, β -catenin–mutated exon 7/8, mixed inflammatory and β -catenin–mutated exon 3, and mixed inflammatory and β -catenin–mutated exon 7/8, representing 8%, 4%, 8%, and 5% of HCAs, respectively (4). Mixed inflammatory and β -catenin–mutated HCAs are primarily inflammatory, as they are histologically more similar to inflammatory HCAs, and *CTNNB1* mutation is believed to be a second genetic event (Fig S2, Fig 7) (5). Additional immunohistochemical staining is required in all cases of suspected inflammatory HCAs to exclude superimposed *CTNNB1* mutations. Similar to “pure” β -catenin–mutated exon 3 HCAs, mixed inflammatory and β -catenin–mutated exon 3 HCAs are associated with higher rates of HCC transformation (4).

β -catenin–mutated HCAs may show an increased number of atypical cells and expression of nuclear β -catenin and/or glutamine synthetase at immunohistochemical analysis. Glutamine synthetase is a downstream target of the wingless-type integration site protein ligand/ β -catenin pathway and shows diffuse staining in some exon 3 mutations (non-S45) but is patchy and faint in exon 7/8 mutations. Glutamine synthetase staining is more sensitive than nuclear β -catenin staining, which often yields negative findings in exon 7/8 mutations (5).

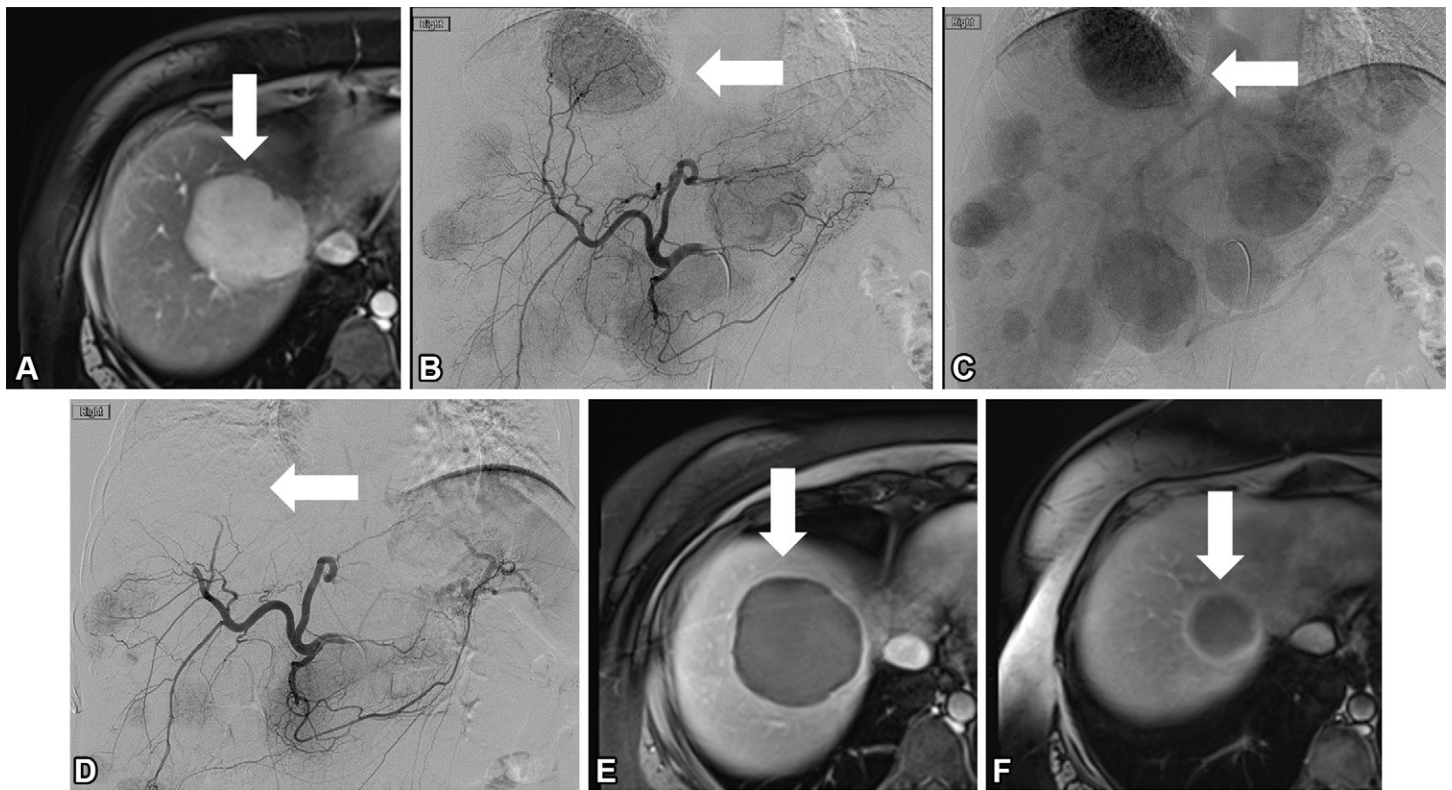
At MRI, β -catenin–mutated exon 3 HCAs may have a T2-hyperintense central scar and show peak enhancement in the arterial phase that may persist to the portal venous phase (14,16,18,20–21). An important distinguishing feature of β -catenin–mutated exon 3 HCAs is that most (>80%) of them are iso- or hyperintense during the hepatobiliary phase (45,46), corresponding to increased OATP1B1/3 expression (Fig 8) (20,22,23,47,48). At the molecular level, increased wingless-type integration site protein ligand/ β -catenin signaling is highly associated with OATP1B1/3 expression and is possibly a downstream target (49,50).

The FDG PET/CT findings of β -catenin–mutated exon 3 HCAs have not yet been reported. Although the US findings are not well established, in two studies (19,25), each involving a single β -catenin–mutated HCA, the tumor was hyperechoic at gray-scale US and had arterial phase hyperenhancement. The HCA in one study (25) reportedly had portal venous washout, while the HCA in the other study (19) was isoechoic.

Sonic Hedgehog HCA

Sonic hedgehog HCAs represent approximately 4% of HCAs and are strongly associated with higher body mass index (32 kg/m²) and longer OC use (median length, 22.5 years). Like β -catenin–mutated exon 7/8 HCAs, these tumors were previously considered to be unclassified until they were identified as a distinct subtype in 2017 (4,7,44). Sonic hedgehog HCAs have the strongest association with background hepatic steatosis, affecting up to 67% of patients, as proven at histopathologic analysis (4). This subtype is characterized by activating mutations of the sonic hedgehog pathway, which is

Figure 7. Mixed inflammatory and β -catenin–mutated exon 3 adenomatosis in a 36-year-old woman with no known androgenic risk factors. (A) Axial gadolinic acid–enhanced T1-weighted fat-suppressed gradient-recalled-echo dynamic MR image during the arterial phase shows a large (>5 cm) hypervascular mass (arrow) in the right hepatic lobe dome. (B, C) Late arterial phase (B) and portal venous phase (C) catheter angiograms of the common hepatic artery show multifocal hypervascular masses (arrow) that continue to be hyperenhanced in the portal venous phase. (D) Catheter angiogram after embolization shows the mass (arrow) with no residual vascular supply. (E, F) Axial portal venous phase T1-weighted fat-suppressed gradient-recalled-echo dynamic MR images show no residual enhancement in the treated mass (arrow) at 2-month (E) and 6-month (F) follow-up, and decreased size of the treatment cavity.



responsible for regeneration and repair in postembryonic tissues (51). At histopathologic analysis, sonic hedgehog HCAs are hypervascular with many isolated veins and arteries, but they do not have the ductular reaction or inflammation seen in inflammatory HCAs. Sonic hedgehog HCAs are associated with histologic bleeding, even when they are smaller than 5 cm, and are the most likely subtype to manifest clinically with symptomatic bleeding (4,5). Figure 9 shows a sonic hedgehog HCA smaller than 5 cm that had bleeding, seen pathologically and radiologically, without symptoms, and Figure 10 shows a sonic hedgehog HCA that had symptomatic bleeding. At initial immunohistochemical analysis, prostaglandin D2 synthase was identified as a specific but insensitive marker. Currently, diffuse or heterogeneous overexpression of arginosuccinate synthase 1, compared with the expression of this enzyme in the background liver, is used to identify the hedgehog HCA subtype (52).

The imaging patterns for this new subtype are not well established. At MRI, sonic hedgehog HCAs may show arterial phase hyperenhancement, with or without venous washout, and internal blood products due to high rates of bleeding (7). Investigators in one study involving four sonic hedgehog HCAs (22) reported that all four tumors were hypointense in the hepatobiliary phase. At histopathologic analysis, three of the tumors had decreased OATP expression, while one had preserved OATP expression.

Unclassified HCA

The remaining 7% of HCAs are unclassified and cannot be categorized into one of the described subtypes (inflammatory, HNF-1 α , β -catenin–mutated exon 3, β -catenin–mutated exon 7/8, mixed inflammatory and β -catenin–mutated exon 3, mixed inflammatory and β -catenin–mutated exon 7/8, sonic hedgehog) on the basis of molecular or immunohistochemical analysis findings. No discerning clinical or histopathologic features have been established for this group. The imaging findings of unclassified HCAs are not well established and are further confounding because they have probably included sonic hedgehog and β -catenin–mutated exon 7/8 HCAs in prior series (14–16,18,20,21). On the basis of the Bourdeaux classification system (29,30), up to 20% of the HCAs in glycogen storage disorders are unclassified. However, no cases of glycogen storage disorders were identified among unclassified HCAs in the 2017 Bourdeaux update (4), suggesting that they were redistributed to other subtypes that were previously considered to be unclassified.

Complications

Bleeding

Bleeding has been described in all HCA subtypes at varying rates and can be life threatening (Figs 1, 2, 5, 8–10). HNF-1 α HCAs are associated with the lowest rate of histologic bleeding

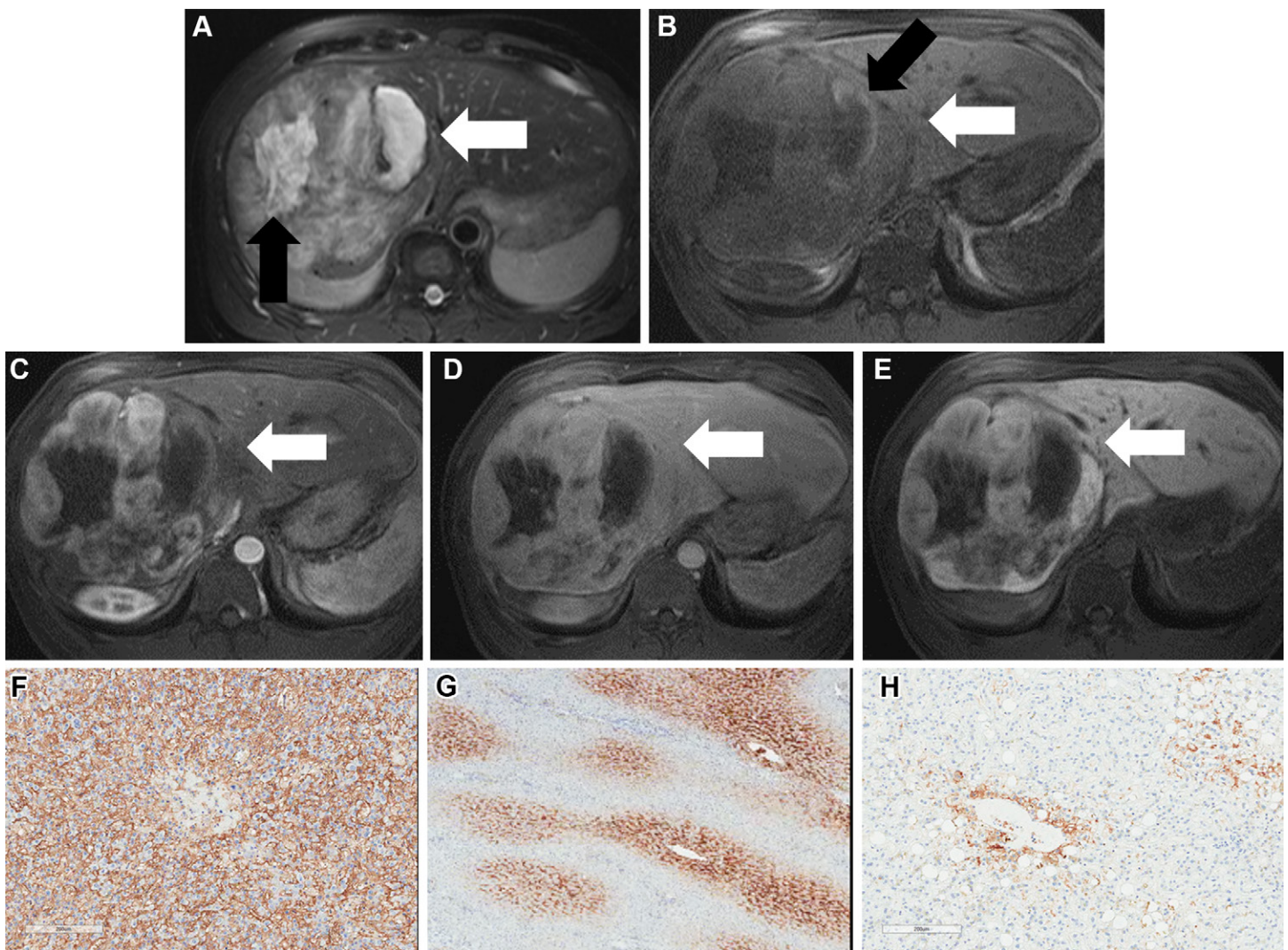


Figure 8. Surgically proven β -catenin–mutated exon 3 HCA in the right hepatic lobe in a 22-year-old man. (A) Axial T2-weighted fat-suppressed MR image shows a mass (white arrow) with a central hyperintense scar (black arrow). (B) Axial gadolinic acid–enhanced T1-weighted fat-suppressed gradient-recalled-echo dynamic MR image shows a mainly isointense mass (white arrow) with a thin area of high signal intensity (black arrow) that corresponded to blood products at histologic analysis. (C, D) Arterial phase MR image (C) shows the mass (arrow) with avid heterogeneous enhancement, which persists on the portal venous phase MR image (D). (E) Hepatobiliary phase MR image shows the mass (arrow) with areas of hyperintensity relative to the background liver. (F) Photomicrograph shows strong diffuse OATP1B1/3 overexpression in the β -catenin–mutated exon 3 HCA. (OATP1B1/3 stain; original magnification, $\times 20$.) (G, H) Photomicrographs of the background liver (G) and HNF-1 α HCA (H) for comparison show markedly reduced OATP1B1/3 expression. (OATP1B1/3 stain; original magnification, $\times 20$.)

(~40%), while β -catenin–mutated exon 7/8 HCAs (>90%), followed by sonic hedgehog HCAs (~80%), involve the highest rates. Histologic bleeding dramatically increases beyond 30 mm. At this cutoff, the rate of bleeding at histology increases from 26% to over 50%. (4). Rates of symptomatic bleeding also vary by subtype, with sonic hedgehog and β -catenin–mutated exon 7/8 HCAs involving the highest rates (~70% and 30%, respectively). The discrepancy between rates of histologic and symptomatic bleeding suggests that some bleeding seen at the histologic level may be subclinical.

Independent risk factors for symptomatic bleeding include HCA subtype (β -catenin–mutated exon 7/8 and sonic hedgehog), chronic alcohol consumption, and larger tumor size (43). HCA size larger than 5 cm traditionally has been considered a strong risk factor for bleeding on the basis of findings in older studies and is the threshold for surgical resection (6,53,54).

Rates of bleeding increase with tumor size, with an odds ratio of 1.54 per 1-cm increase (55). However, investigators in a relatively recent large cohort study (43) found that the β -catenin–mutated exon 7/8 and sonic hedgehog HCA subtypes and chronic alcohol consumption were much more strongly associated with bleeding than tumor size, suggesting that the traditionally used size threshold alone may be insufficient in risk stratification. In a smaller study (56), exophytic growth was found to be associated with bleeding, but this result could not be reproduced in the larger, more recent study (43).

Historically, it has been believed that inflammatory HCAs are associated with the highest risk of bleeding. This belief was based on initial experiences published by the Bordeaux group and the 2006 classification system (2). According to the 2006 classification, hemorrhage seen at histologic analysis was most strongly associated with the inflammatory subtype,

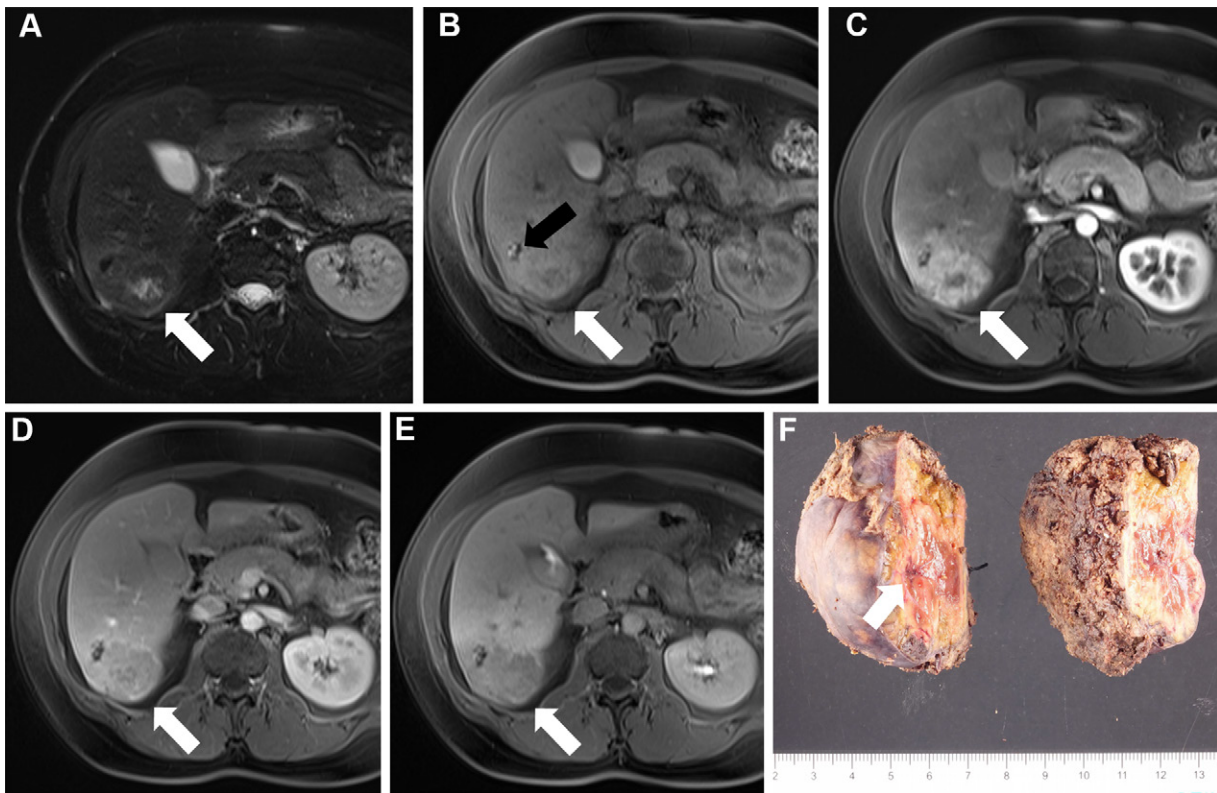


Figure 9. Sonic hedgehog HCA in the right hepatic lobe in a 35-year-old woman. (A) Axial T2-weighted fat-suppressed MR image shows a mass (arrow), a sonic hedgehog HCA, in the right hepatic lobe. (B) Axial gadolinic acid-enhanced T1-weighted fat-suppressed gradient-recalled-echo dynamic MR image shows a mostly isointense mass (white arrow) and a peripheral area of intrinsic high signal intensity with susceptibility artifact (black arrow) during the unenhanced phase, corresponding to hemorrhage seen at pathologic analysis. (C, D) T1-weighted MR images show the mass (arrow) with avid enhancement during the arterial phase (C) and washout during the portal venous phase (D). (E) On the hepatobiliary phase T1-weighted MR image, the HCA (arrow) is hypointense compared with the background liver. (F) Gross pathologic specimen shows an encapsulated mass with blood products (arrow).



Figure 10. Ruptured sonic hedgehog HCA in a 22-year-old woman. (A, B) Axial arterial phase (A) and coronal portal venous phase (B) CT images show a subcapsular mass (white arrow in A) in the left hepatic lobe, with areas of active extravasation (black arrow) that increased during the portal venous phase and a subtle area of hyperenhancement (white arrow in B) corresponding to a second HCA. (C) Emergent catheter angiogram shows a large torrential area of active extravasation (black arrow) and the second HCA (white arrow).

although the overall sample size included only 37 inflammatory HCAs and seven unclassified HCAs. Since then, studies with larger sample sizes and including multivariable statistical analyses have found that the newly described β -catenin–mutated exon 7/8 and sonic hedgehog subtypes (43) are stronger risk factors. Furthermore, newer studies distinguish histologic bleeding from symptomatic bleeding, the latter of which may be more clinically relevant.

Malignant Transformation

The overall rate of malignant transformation among surgical cohorts is 4.2% (57). Most HCCs arising from HCAs are well differentiated (5) and have a better prognosis than that for HCCs arising from noncirrhotic livers (58).

β -catenin–mutated exon 3 HCA subtype, tumor size larger than 5 cm, male sex, and chronic liver disease (specifically, glycogen storage disorders and vascular liver disease) are

established risk factors for HCC transformation (5). Transformation to HCC of varying rates has been described with all HCA subtypes. β -catenin–mutated exon 3 HCA is associated with the highest risk of transformation to HCC by far (>40%), while the risk is negligible among β -catenin–mutated exon 7/8 HCAs (4). Malignant transformation among β -catenin–mutated exon 3 HCAs appears to take at least 10 years (59). Risk of HCC transformation increases with tumor size but has been reported in HCAs smaller than 5 cm (57). HCAs in men have a higher risk for malignant transformation, regardless of subtype. The overall risk is 10 times higher in men than in women (53). In one study (4), among men, malignant transformation occurred in 41% of cases of non–at-risk HCA subtypes and in 61% of cases of at-risk HCA subtypes, the latter group defined as sonic hedgehog and β -catenin–mutated exon 3 HCAs (4). Thus, for men with lesions of any size suspected of being HCA, surgical resection is recommended (6). The molecular mutations, risk factors, complications, and key imaging findings associated with the different HCA subtypes are summarized in the Table.

Management

Initial Diagnosis

All patients suspected of having HCA should undergo contrast-enhanced MRI (6). MRI can readily depict the most common HCA subtypes, which account for 70%–80% of these tumors. For example, the presence of microscopic fat has a sensitivity of 95% and specificity of 85% for HNF-1 α HCAs (18). Moderate T2 hyperintensity and persistent enhancement in the venous phase have a sensitivity of 85%–88% and specificity of 88%–100% for inflammatory HCAs (6). The atoll sign has fair sensitivity (60%) but high specificity (97%) for inflammatory HCAs. A T2-hyperintense central scar has a sensitivity of 80% and specificity of 91% for β -catenin–mutated exon 3 HCAs (18).

MRI can be performed with extracellular or hepatobiliary contrast agents (gadobenate dimeglumine or gadoxetic acid). HCA subtypes show varying degrees of signal intensity in the hepatobiliary phase, which can improve diagnostic performance. For example, HNF-1 α HCAs have the lowest signal intensity during the hepatobiliary phase (18,20–22), and the inclusion of this criterion improves specificity to 96% (18). HCAs generally are hypointense on hepatobiliary phase MR images; the exceptions to this feature are β -catenin–mutated exon 3 subtypes and up to 30% of inflammatory HCAs, which may be iso- to hyperintense during the hepatobiliary phase (10,18,20–22,45,46).

Distinguishing inflammatory HCAs with hepatobiliary phase uptake from FNHs can be challenging, although quantitative MRI assessment of the hepatobiliary phase with both gadobenate dimeglumine and gadoxetic acid has been shown to have strong discriminatory power, with FNHs having significantly higher signal intensity ratios (60,61). Other ancillary features such as a lobulated morphology and a central scar are more common in FNHs (61). Both FNHs and HCAs can restrict diffusion; however, in one study (62), the mean apparent diffusion coefficient (ADC) value of HCAs was

lower than that of FNHs; no differences in ADC values were identified among the HCA subtypes themselves.

Rarer HCA subtypes such as sonic hedgehog and β -catenin–mutated exon 7/8, which are associated with the highest bleeding risk, may be challenging to diagnose by using imaging alone, as they have no specific imaging findings. When the diagnosis or a definitively indolent subtype cannot be established with imaging, biopsy may be necessary (43,60). Ultimately, there is no consensus regarding specific indications for biopsy, and lesions suspicious for HCA may require a customized multidisciplinary discussion at a liver referral center (63).

Initial Management

When HCA is diagnosed in men, surgical resection is uniformly recommended because of the substantially higher risk of malignancy, regardless of tumor subtype and size. It is also recommended that patients with β -catenin–mutated exon 3 HCA undergo surgical resection, regardless of tumor size, owing to the higher risk of malignancy (6). For women, risk factor reduction measures such as cessation of OC use and weight loss are recommended first, followed by repeat MRI for assessment of size change. If the HCA grows (20% diameter increase, adopted from RECIST [Response Evaluation Criteria in Solid Tumors] criteria for “progression”) or remains larger than 5 cm, surgical resection is recommended owing to the increased risk of bleeding and malignant transformation. Transarterial embolization and ablation are considered second-line treatment alternatives according to European Association for the Study of the Liver guidelines and are reserved for patients who are not surgical candidates (6). However, recent studies have found success with use of local-regional treatment only and are discussed later.

Surveillance

If the HCA has a stable or small size and has remained smaller than 5 cm for a considerable period, conservative management with annual surveillance imaging is recommended. MRI is generally used for serial follow-up examinations (37). Unlike CT, MRI does not involve the use of ionizing radiation, and most patients with HCAs are relatively young individuals who are expected to undergo multiple imaging examinations. Furthermore, the criterion for progression is adopted from RECIST, in which axial CT or MRI is used for size measurements (6). Nevertheless, follow-up with US has been proposed as an alternative in select circumstances, as it may be more cost effective (6). It is unclear whether all follow-up MRI examinations require hepatobiliary contrast agents. Hepatobiliary agents may be more sensitive for detecting small new lesions and theoretically may be used to detect gain of β -catenin mutations in an inflammatory adenoma, which is presumed to reflect a second genetic event (5).

The length and frequency of follow-up (eg, annual versus biannual) are not well established, although the most stringent follow-up advocated by one group involves annual follow-up until menopause in women (60). In one study of postmenopausal patients with HCAs, 44% of tumors were undetectable, 20% of them had shrunk, and 33% of them were

Key Histopathologic, Clinical-Demographic, and Imaging Features of HCA Subtypes

Subtype	Gene Mutation(s)	Risk Factors	Hemorrhage Risk (%)	Malignancy Risk (%)	Key Imaging Features
Inflammatory	<i>IL6ST</i> , <i>FRK</i> , <i>JAK1</i> , <i>STAT3</i> , or <i>GNAS</i>	OC use, 20.0 y; F:M ratio, 98:2; obesity; alcohol use	13	6	Atoll sign, moderate T2 hyperintensity, arterial phase hyperenhancement, venous phase hyperenhancement, varying hepatobiliary phase signal intensity (mostly hypointensity but iso- to hyperintensity in up to 30% of cases), background fatty liver
HNF-1 α	Hepatocyte nuclear factor 1 α	OC use, 12.0 y; F:M ratio, 90:10; germline mutation	18	5	Hyperechoic (due to microscopic fat), T1 opposed-phase signal loss (due to microscopic fat), arterial phase hyperenhancement, venous phase washout, hepatobiliary phase hypointensity (lowest signal intensity ratio), FDG avidity
β -catenin–mutated exon 3	<i>CTNNB1</i> exon 3	OC use, 8.0 y; F:M ratio, 60:40	7	47	Central scar, arterial phase hyperenhancement, hepatobiliary phase iso- to hyperintensity
Mixed inflammatory and β -catenin–mutated exon 3	<i>CTNNB1</i> exon 3 and <i>IL6ST</i> , <i>FRK</i> , <i>JAK1</i> , <i>STAT3</i> , or <i>GNAS</i>	OC use, 12.5 y; F:M ratio, 70:30	4	27	Arterial phase hyperenhancement, hepatobiliary phase iso- to hyperintensity
β -catenin–mutated exon 7/8	<i>CTNNB1</i> exon 7/8	OC use, 9.5 years; F:M ratio, 100:0	33	0	Generally no specific imaging features; arterial phase hyperenhancement and/or hepatobiliary phase hypointensity may be seen
Mixed inflammatory and β -catenin–mutated exon 7/8	<i>CTNNB1</i> exon 7/8 and <i>IL6ST</i> , <i>FRK</i> , <i>JAK1</i> , <i>STAT3</i> , or <i>GNAS</i>	OC use, 22.0 y; F:M ratio, 76:24; obesity; alcohol use	5	5	Generally no specific imaging features; arterial phase hyperenhancement and/or hepatobiliary phase hypointensity may be seen
Sonic hedgehog	<i>INHBE</i> and <i>GLI1</i>	OC use, 22.5 y; F:M ratio, 100:0; obesity	71	6	Generally no specific imaging features; blood products and/or hepatobiliary phase hypointensity may be seen
Unclassified	None established	OC use, 10.5 y; F:M ratio, 81:19	11	14	Generally no specific imaging features; arterial phase hyperenhancement may be seen

Note.— Hemorrhage risk refers to symptomatic bleeding. Malignancy risk refers to prevalence of malignancy at histopathologic analysis. F:M = female-to-male.

stable at follow-up (64). The length of the follow-up period and frequency of follow-up examinations also may depend on the HCA subtype. HNF-1 α HCAs are twice as likely (40%) as inflammatory HCAs to progress during the imaging surveillance period (37). Overall, more evidence-based information is needed regarding the modality, contrast agent, frequency of examinations, and overall length for follow-up of HCAs.

Local-Regional and Surgical Treatment

Treatment of ruptured or actively bleeding HCAs typically involves fluid resuscitation followed by transarterial embolization to achieve hemodynamic stability (Fig 10) (43,65,66). In rare cases, laparotomy with packing may be necessary to control the bleeding. Immediate surgical resection of the HCA is associated with increased morbidity and thus is typically delayed for several months. In the interim, the blood products may resorb, the inflammation may decrease, and the culprit HCA may be easier to identify. Some patients with bleeding HCAs can be treated with transarterial embolization only, without delayed surgical resection, particularly if the HCAs shrink or no longer enhance (43,65–68). Investigators in one relatively recent study (66) found that complete ne-

crosis could be achieved in 12 of 18 bleeding HCAs by using transarterial embolization. At follow-up, the authors proposed conservative management for those women whose HCAs had reduced to smaller than 5 cm or had no viable tissue (66); in such cases, surgery can be avoided in up to 40% of patients with use of this approach. Men would still require surgical resection because of the high risk of HCC transformation; if and how transarterial embolization affects malignancy risk are still unknown.

For elective treatment of growing HCAs or HCAs that remain larger than 5 cm, guidelines of the European Association for the Study of the Liver recommend surgical resection. Surgical resection can be performed with a laparoscopic or open technique (65). Large HCAs or adenomatosis may require a staged approach (Fig S3). Liver transplant is rarely necessary and is reserved for patients with malignant transformation or those with chronic liver disease such as glycogen storage disease or vascular liver disease (43). Liver adenomatosis by itself (without chronic liver disease) is not an indication for transplant (65).

While transarterial embolization is typically performed in the emergent setting, it is increasingly being used in elective

settings. Thus, indications have expanded beyond stabilization of active bleeding and include treatment of surgically unresectable HCAs, adenomatosis with symptoms, patient's wish for pregnancy or ongoing pregnancy, reduced tumor size and vascularity as a bridge to surgery, or alternative to surgery altogether (Fig 7) (68–70). Transarterial embolization is safe overall, with rare complications, which are similar to those associated with HCC embolization and include bleeding, abscess, and thrombus.

Transarterial embolization is also effective for reducing the size of HCAs, although patients may require multiple sessions. In a systematic review (70), 75% of HCAs were found to have a greater than 30% decrease in diameter and 10% of HCAs completely regressed. Investigators in an observation study (68) reported that 27 of 28 (96%) HCAs that were larger than 5 cm before transarterial embolization shrunk to smaller than 5 cm, which is below the threshold for surgical resection, after the embolization.

Thermal ablation is another local-regional option for treatment of HCAs (63). It can be performed with either radiofrequency ablation or microwave ablation (Fig S4) and is typically used in patients with HCAs smaller than 5 cm (71–73). However, HCAs smaller than 5 cm in women are usually managed with surveillance rather than surgical resection, unless they grow. Thus, indications for ablation can include patient preference for intervention rather than surveillance, multifocal HCAs or adenomatosis, pregnancy wish or ongoing pregnancy, patients who cannot stop OC use (thus, the HCA would be expected to grow), and a small HCA deemed to be surgically nonresectable but requiring treatment (eg, male patient, β -catenin–mutated exon 3) (70,71). Thermal ablation is safe and effective overall, although patients may require more than one ablation session. Major complications are uncommon and can include liver abscess and bleeding (71).

Pregnancy

Current guidelines recommend periodic sonographic monitoring for tumor growth every 6–12 weeks for HCAs of any size in pregnant women (6). For small (<5 cm) HCAs, the risk is minimal, although about 25% of these tumors will grow during pregnancy (74). Another 25% will regress, and the rest will remain stable. Bleeding during pregnancy has been reported in only HCAs 6.5 cm or larger, mostly during the third trimester, and it has not been reported in the first trimester (75). The true rate of bleeding during pregnancy is unknown, given selection bias and the probable underreporting of asymptomatic HCAs during uncomplicated pregnancies. However, when bleeding occurs, it potentially can be fatal to both the mother and the fetus (43,75). Treatment options include transarterial embolization or surgery, which would require multidisciplinary evaluation by surgeons, interventional radiologists, and obstetricians, particularly when evaluating risk to both the fetus and the mother (6,43,75).

Conclusion

New research on the genetic underpinnings of HCAs in the past decade have significantly improved our understanding of

the biologic features of these tumors. These unique molecular pathways affect the imaging appearances and prognoses associated with HCAs, which in turn guide management. Multiphasic MRI with hepatobiliary contrast agents remains the best method to noninvasively detect, characterize, and monitor HCAs. Nevertheless, it is important to understand their appearance at imaging with other modalities as well. HCAs may mimic other neoplasms or require imaging with other modalities according to the clinical setting, such as pregnancy or suspected rupture. Bleeding and malignant transformation are the two major complications and depend on the size, clinical-demographic features, and subtype of the HCA. These complications are the basis for surgical resection of HCAs larger than 5 cm, growing HCAs, HCAs of any size among men, and β -catenin–mutated HCAs of any size. Promising emerging data and results from transarterial embolization and thermal ablation suggest that these techniques may be alternatives to surgical resection.

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